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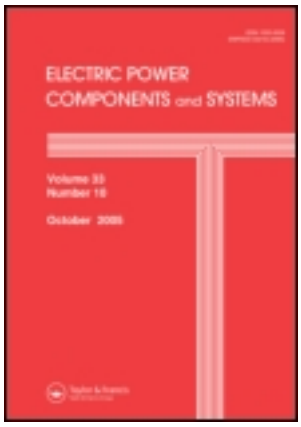

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An Assessment of the Methods for Calculating Ampacity of Underground Power Cables

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An Assessment of the Methods for Calculating Ampacity of Underground Power Cables

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Endurance of an insulation material to high temperatures determines the maximum current-carrying capacity (ampacity) of an underground power cable. Cable ampacity is calculated conventionally using the installation conditions and maximum steady-state operation temperature according to IEC-60287 standard. In this article, finite element method results are compared with IEC-60287. A further verification has also been made with experiments. In this work, ampacity analyzes of 154 kV high voltage XLPE underground power cable are made using ANSYS 5.61 finite element analysis software. An experimental set-up is developed to measure conductor and surface temperature of the cable in underground conditions. The results of experiments and numerical calculations are compared to the IEC-60287 standard. Additionally, the thermal regions of three cables in flat installation are investigated to show the versatility of the finite element method. The effects of insulation thickness and external thermal source on ampacity are also analyzed using the finite element method.

Keywords ampacity, finite element, thermal rating, underground power cables

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Introduction

Underground power cables are more expensive to install and maintain than overhead lines. The greater cost of underground installation reflects the high cost of materials, equipment, labor, and time necessary to manufacture and install the cable. The large capital cost investment makes it necessary to use their full capacity. On the other hand, conductor temperature of a power cable limits its ampacity (i.e., maximum allowable temperature for XLPE insulated cables is 90°C). Also, the operating temperature adversely affects the useful working life of a cable. Excessive conductor temperature may irreversibly damage the cable insulation and jacket.

A successful model was proposed for calculating ampacity of underground cable by Neher-McGrath in 1957 [1]. The Neher-McGrath Model has been widely accepted for over 40 years. Today, the greater majority of utilities and cable manufacturers have been using the IEC-60287 standard [2]. This method employs a lot of simplifications and has its limitations. Thus, it is not reliable for the analysis of complex configurations.

The finite element method (FEM) is more powerful and precise in terms of geometrical modeling complexity. The finite element method solves problems that are described by partial differential equations, using numerical techniques. A domain to be analyzed is represented as an assembly of finite elements. Approximating functions in finite elements are defined in terms of nodal values of a physical field. A continuous physical problem is transformed into a discretized finite element problem with unknown nodal values. For linear problems, developed system of linear algebraic equations are solved numerically. Values inside finite elements can be obtained using nodal solutions.

Although the idea of dividing a continuum into small finite pieces had been first suggested by Courant in 1943 [3], the development of the finite element method coincided with major advances in computers technology and programming languages. Under operation circumstances, the measurement of the conductor temperature of the HV underground cable is very difficult. Thus, the usual conditions of installation and operation can be simulated in laboratory environment. A heating test bed according to IEC62067, which is an extension of IEC60840 for higher voltages, is set up to measure temperature distribution. The cable loaded with various currents can be investigated. Consequently, the thermal rating of the cable can be determined.

Ampacity analysis of cables with FEM has been studied by many researchers [4–7]. Lately, an IEC technical report, namely IEC TR 62095 [8], has been published for current ratings calculation using the finite element method when IEC 60287 [2] cannot be applied. Nahman and Tanaskovic [9] have measured and modeled thermal regions outside the cable using FEM. Even if they used linear triangular elements in the FEM analysis, their results were in good agreement with their measurements. If they had used quadratic quadrilateral elements in their method, improved results could be obtained.

In this project, current rating calculations for power cables in steady-state conditions were performed using IEC 60287, finite element method and experiments. The models developed for the finite element analysis were verified with IEC 60287 and experiments. A numerically efficient finite element model is obtained with sensitivity analysis. Since the domain under the cable is not an isotherm, the control volume to be analyzed is very effective in the degree of accuracy of the analysis. Fundamental terms and boundary conditions in the FEM analysis were clearly identified in the results section, which can help engineers to develop their models for ratings calculations.

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Ampacity Calculation

The ampacity calculation of a power cable using IEC 60287 formulation can be written in the following form:

$$I = \sqrt{\frac{\theta_c - \theta_a - \theta_d}{(T_1 + (1 + \lambda_1)(T_3 + T_4))R_{ac}}} \tag{1}$$

where

θ_c	maximum operating conductor temperature °K
θ_a	ambient temperature °K
θ_d	temperature rise due to dielectric loss °K
λ_1	sheath loss factor
R_{ac}	AC resistance of conductor at temperature θ_c

T_1, T_3, T_4 thermal resistances of insulation, covering and soil (°Km/W), respectively. The external thermal resistance T_4, depends on the diameter of the cable, the depth of laying, the thermal characteristics of the soil, mode of installation and on temperature rise generated by neighboring cables.

Finite Element Method

A basic equation of heat transfer for an isotropic body with temperature dependent heat transfer has the following form:

$$-\left(\frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y}\right) + Q = \rho c \frac{\partial \theta}{\partial t} \tag{2}$$

where q_x and q_y are components of heat flow through the unit area; Q is the inner heat generation rate; q is density; c is thermal heat capacity; θ is temperature, and t is time.

Fourier's law describes the heat flow equations as

$$q_x = -k\frac{\partial \theta}{\partial x}, \quad q_y = -k\frac{\partial \theta}{\partial y} \tag{3}$$

Here k denotes the thermal coefficient of the material. Substitution of the above relations gives

$$\frac{\partial}{\partial x}\left(k\frac{\partial \theta}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial \theta}{\partial y}\right) + Q = \rho c \frac{\partial \theta}{\partial t} \tag{4}$$

Then the variation of the temperature and temperature gradients inside an element can be expressed in terms of nodal temperatures using shape functions N_i as

$$\{\theta\} = [N]\{\theta^e\} \tag{5}$$

Differentiation of the temperature field gives

$$\left[\frac{\partial\theta}{\partial x}\right] = \left[\frac{\partial N}{\partial x}\right]\{\theta^e\} \tag{6}$$

where θ^e are the nodal temperatures for the eth element. N represents the quadratic shape functions for 8-node quadrilateral elements (serendipity elements)

While shape functions are expressed through the local coordinates ξ, η, the matrix contains derivatives in respect to the global coordinates x, y. Derivatives can be easily converted from one coordinate system to the other by means of the chain rule of partial differentiation:

$$\begin{bmatrix} \dfrac{\partial N_i}{\partial \xi} \\ \dfrac{\partial N_i}{\partial \eta} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial x}{\partial \xi} & \dfrac{\partial x}{\partial \eta} \\ \dfrac{\partial y}{\partial \xi} & \dfrac{\partial y}{\partial \eta} \end{bmatrix} \tag{7}$$

Using Galerkin method, we can rewrite Eq. (4) in the following form:

$$\Pi = \iint \left(\begin{array}{c} \dfrac{1}{2}\{\theta_n\}^T k \left(\dfrac{\partial^2[N]^T}{\partial x^2}\dfrac{\partial^2[N]}{\partial x^2} + \dfrac{\partial^2[N]^T}{\partial y^2}\dfrac{\partial^2[N]}{\partial y^2} \right)\{\theta_n\} - \\ \{\theta_n\}^T[N]^T q \end{array} \right) dxdy \tag{8}$$

Equations are rearranged to give stiffness matrix of $[K]$ and force vector of $\{f\}$ as

$$[K] = \iint \left(\frac{\partial^2[N]^T}{\partial x^2}\frac{\partial^2[N]}{\partial x^2} + \frac{\partial^2[N]^T}{\partial y^2}\frac{\partial^2[N]}{\partial y^2} \right) kdxdy \tag{9}$$

$$\{f\} = \iint [N]^T q dxdy \tag{10}$$

FEM equations are found by minimization of functional Π in terms of nodal temperatures.

$$\left\{\frac{\partial \Pi}{\partial \theta_n}\right\} = 0 \quad \text{and} \quad [K]\{\theta_n\} = \{f\} \tag{11}$$

Stiffness matrix $[K]$ and force vector $\{f\}$ integrals are calculated on each element numerically (using Gaussian quadrature), and then total stiffness matrix is set up by summation of each equation system. The total equation system is then solved for each node.

Numerical and Experimental Work

154 kV underground cable in single installation is analyzed using FEM and IEC-60287. These results are used for determining the insulation thickness representing the soil's thermal resistance in a thermal rating experiment. Further further loading cycles are then applied to the test cable, and the values obtained from a full heating of a cycle is compared with FEM and IEC 60287 results. The effects of insulation thickness on current ratings

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were also analyzed using both methods. Another current rating calculation defined in IEC 60287, three cables in flat formation, were also analyzed and compared with FEM analysis. A further application of this problem in the case of a heat pipe installed parallel to the cable was studied as well.

154 kV Single Underground Cable Analysis with FEM and IEC

The ampacity of the underground cable, whose sectional view is illustrated in Figure 1, is analyzed using the finite element method. Under the operation conditions, the amount of heat generated from the cable should be calculated to determine ampacity. Since the limiting operation temperature of XLPE cables is 90°C, the heat generated from the charged cable should be transferred to the environment to reside under this temperature. That's why the correct calculation of heat transfer from the cable affects the ampacity analysis directly. The soil temperature is 20°C for northern Turkey. 1.2 m is the depth the underground cables are normally buried under. However, the temperature and thermal properties of soil depend on various parameters like the moisture content, the seasonal variations and cable depth. These were reported by Brakelmann [10]. In steady-state operation conditions, moisture around the cable can be ignored since the heat generation from the cable dry out the moisture nearby. IEC-60287 standard assumes that the properties of soil and boundary conditions for the calculations are constant. Using these conditions, and assuming the soil is homogenous, heat transfer and ampacity are calculated both by IEC-60287 and the finite element method. The boundary conditions for FEM analysis is shown in Figure 2.

The steady-state calculation of the heat transfer gives general operating ampere capacity. For shorter times, this current can be exceeded under cable manufacturer's limiting values. In this analysis, transient effects are not considered, thus time is not a parameter.

Since the heat transfer coefficients of XLPE and the semi-conductor material are very close to each other, they are assumed to be same and are considered as a single layer. The effects of coating layers and very thin metallic screens are omitted as their thermal resistivities are very small and have very little effect on the results. Generalized

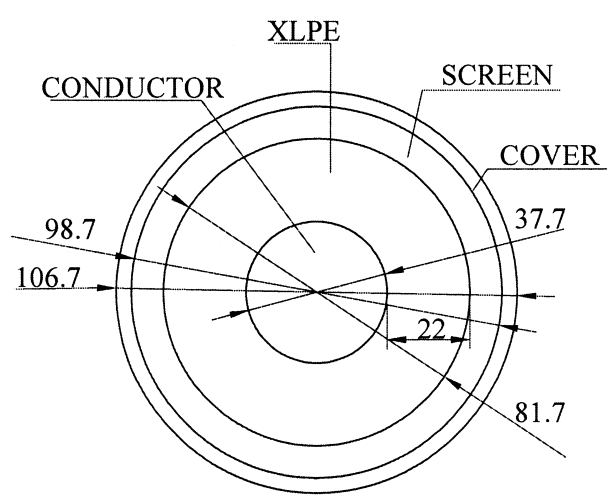

Figure 1. Sectional view of the cable.

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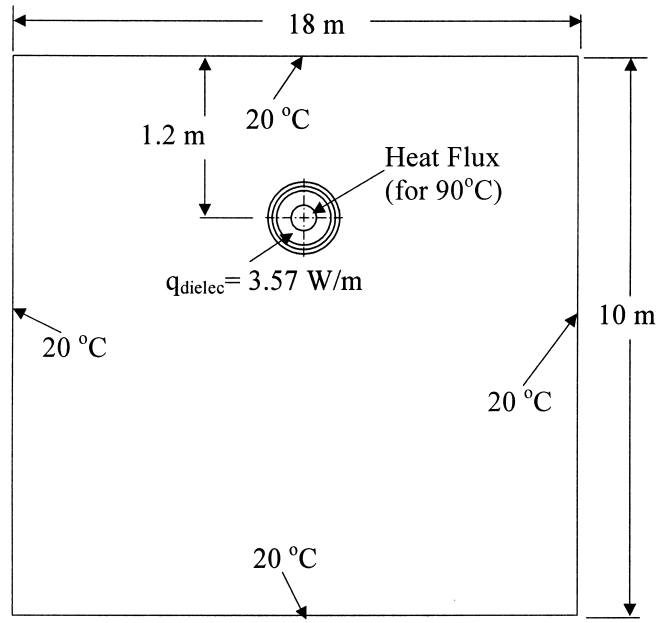

Figure 2. The layout position of the 154 kV cable is given above.

layer dimensions of a 154 kV underground cable and the dielectric losses generated on XLPE are as follows:

$D_{Conductor} = 37.7$ mm
$D_{XLPE+Semi\ Conductor} = 81.7$ mm
$D_{Screen} = 98.7$ mm
$D_{Cover} = 106.7$ mm
$h_{depth} = 1200$ mm
$q_{dielectric} = 3.57$ W/m
$k_{soil} = 1.2$ W/Km
$k_{XLPE} = 0.2857$ W/Km
$k_{screen} = 384.6$ W/Km

For the underground cables, the earth surface can be assumed as isotherm as described in IEC 62095. The domain under the cable is earth, which is an infinite medium. The soil temperature also increases gradually as it gets deeper underneath the earth's surface. However, the soil temperature below 5 meters converges to a certain value and does not fluctuate seasonally [10]. That's why earth underneath the cable can also be taken as isotherm at a certain depth. This effective control volume can be found using sensitivity analysis. Various control volume sizes are analyzed, and heat transfer rates for the half of the region are shown in Figure 3. At the depth of approximately 10 meters, heat transfer rate converges to a value within the 1% range. Heat dissipation increases by 0.6% if the region to be discretized is increased from 5m to 10 m in depth. Thus, a control volume of 10 m depth and 18 m width is taken for all other cases as well.

The insulation thickness of a cable affects the permissible maximum current and heat transfer from the cable. Higher insulation thickness also blocks the heat generated in the conductor. Various insulation thicknesses for a 154 kV underground cable are analyzed

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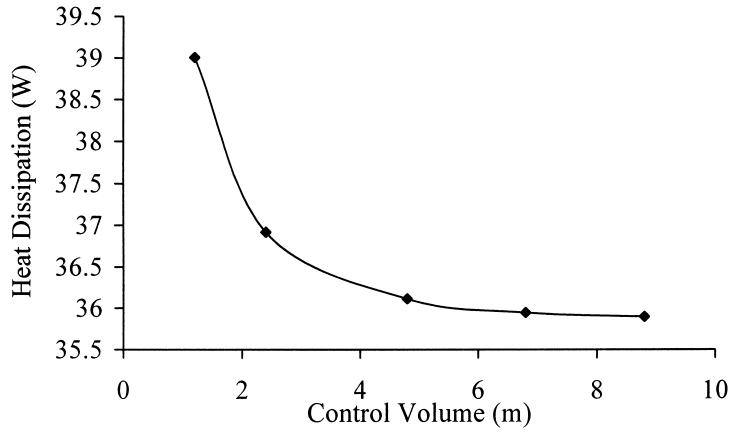

Figure 3. Sensitivity analysis between the control volume depth and heat dissipation.

in this example. For each cable, the heat flux is applied on the conductor found by an iterative approach. The heat flux is increased gradually until a maximum temperature of 90°C on the conductor is achieved. For the cable with 22 mm XLPE insulation, the thermal flux generated on the conductor surface is found to be 0.585 W/m^2.

Since the model is symmetrical about y-axis, only half of the domain is modeled. As a general rule of finite element method, element shapes are kept close to an aspect ratio of unity. Heat generation by dielectric losses is applied on the XLPE domain. 1074 elements are found to be dense enough for the final calculation. FEM analyzes with different mesh densities, such as the number of elements 557, 690, 1074, and 1486, have shown that the total heat transfer on the conductor converges to a certain value by increasing mesh density. This sensitivity analysis regarding mesh density is necessary for evaluating the correctness of the FEM analysis as described in IEC TR 62095. The mesh chosen after the sensitivity analysis are shown in Figures 4 and 5.

Figure 6 show that the ampacity for the cable has a difference of around 1% between the FEM and IEC 60287 models. The results also illustrated that the reduction in the insulation thickness by 22% (from 22 to 17 mm) has an effect of 2.9% increase in the ampacity. Since the heat transfer mechanism is conduction, and this is directly proportional to the contacting surface area, reduction in the diameter reduce both thermal barrier and contacting surface.

Thermal Rating of 154 kV XLPE Power Cable

The experiment conditions are devised to comply with IEC-62067, which is an extension of IEC-60840 for higher voltages section 12.4.7 on power cable systems [11, 12]. Thermal rating standards involve a series of heating cycles by conductor current and cooling cycles with natural cooling. In a heating cycle range from approximately 310 K to 380 K, conductor and shield temperatures measured on experiments can be compared to the FEM and IEC-60287 results to observe the performance of the both methods.

For this purpose, a 15 m of 154 kV XLPE power cable is used in the experiments. A heating system is devised using thermocouples and thermal insulators for the temperature measurements on the conductor and shield of the cable. Thermal resistance of the soil is represented with a paper layer which has a thermal conductivity of 0.475 W/Km.

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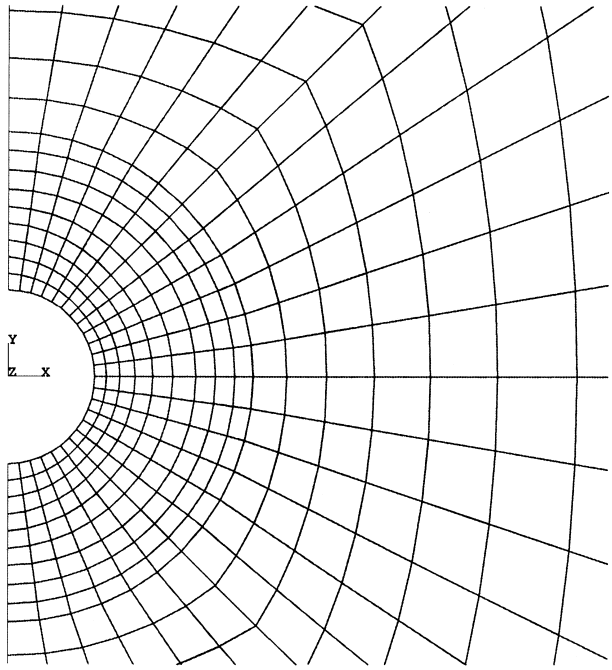

Figure 4. FEM mesh around the cable.

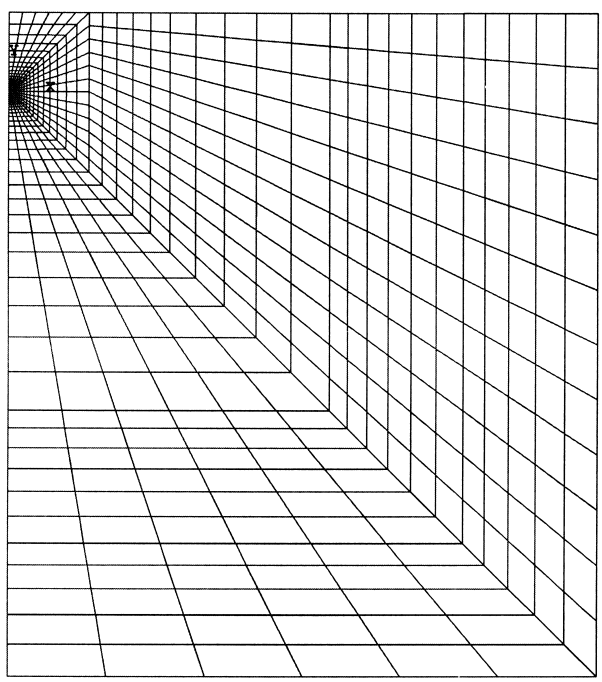

Figure 5. The mesh modeling both the cable and the surrounding soil with 1074 elements.

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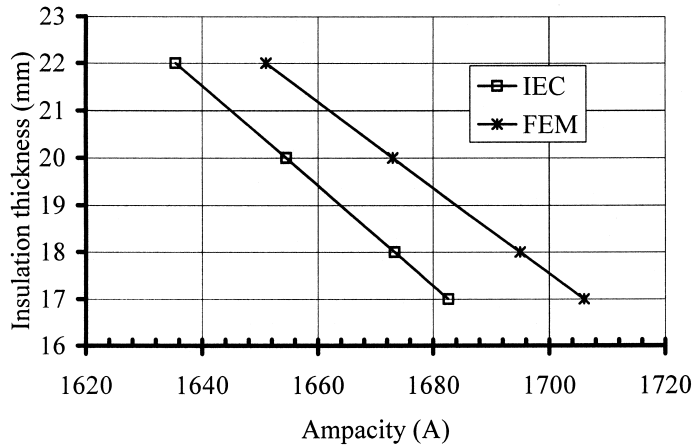

Figure 6. The ampacities for different insulation thickness by both FEM and IEC 60287 model.

The thickness of the paper layer wrapped on the cable is found by a trial and error technique. In steady-state conditions at 363 K conductor temperature, the shield temperature was measured and compared to the analytical values while the paper thickness increased gradually. As the same values of conductor and shield temperatures are obtained from both analytical calculations and experiments, the wrapped paper thickness gives the thermal resistance of the soil. Three thermocouples are used to measure conductor temperature. One is located at the center and two others are located at a distance of one meter from the ends. A precise temperature measurement is achieved by drilling the cables till 3 mm in conductors, as shown in Figure 7. The space at the tip of the thermocouple and the conductor is filled with a highly conductive material.

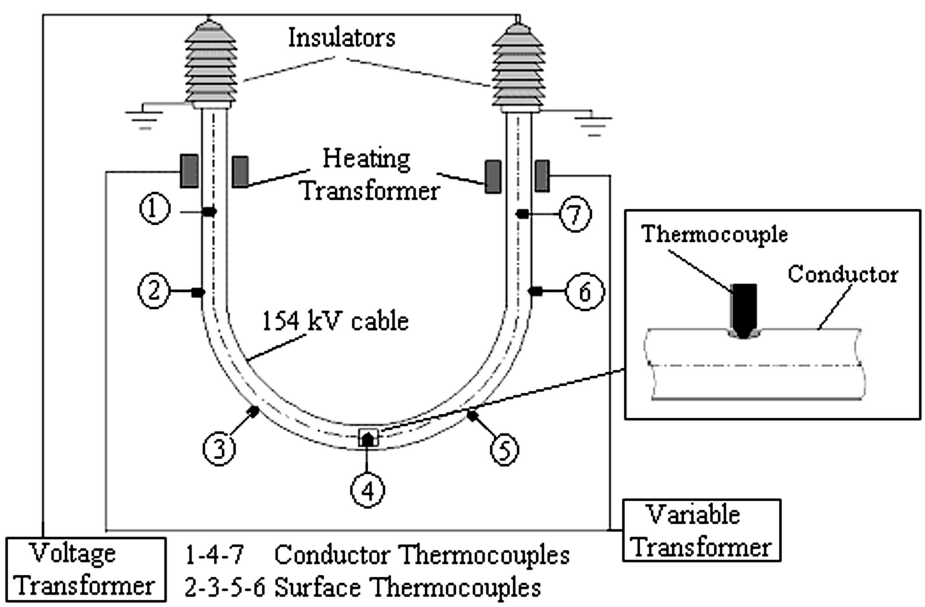

Figure 7. Experimental layout.

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Table 1
Experimental results

Cycle 10		Cycle 11	
T-conductor (°K)	T-shield (°K)	T-conductor (°K)	T-shield (°K)
319	298	327	304
321	300	330	307
335	313	339	316
351	322	355	324
369	330	365	329
381	337	375	332
382	338	377	333
382	338.5	377	333
		377	333

Four thermocouples are positioned evenly in length on the outer surface of the cable. All of the thermocouples are connected to a temperature recorder (Siemens-Kompensograph C1015).

Both ends of the cable are connected to secondary probes of the voltage transformer. Then the cable is placed as the secondary winding of a heating transformer. The cable is gradually heated up in steps and awaited in each step until the difference between each conductor thermocouples are less than 1°K. Each experimental cycle took at least 8 hours to reach maximum temperature. Twenty experiments are performed consequently. The experimental results in 10th and 11th cycles, when stable periodic conditions were achieved, are given in Table 1.

The results from two cycles construct a continuous path, even if the recorded conductor temperatures are different in each experiment. The heat dissipation from the cable is calculated from the measured conductor and shield temperatures.

While the experimental setup was calibrated with analytical calculations at 363 K, the heating experiments ranged from 320 K to 383 K. Consequently the comparison of the FEM and IEC-60287 with experiments gives an indication about both methods' performances. The currents necessary to generate the heat versus conductor temperature are plotted on Figure 8. There is a difference between the FEM and experiment results at lower temperatures. Since the speed of the heating is too high at the beginning of the experiment, the converged temperature values have not been reached. However on the rest of the graph, experimental results are very close to FEM results.

At the beginning of the experiment, the current was increased so much that calculations under steady-state conditions would not be expected to conform. The heating process after 340°K became more stable and the ratio changed linearly. FEM and IEC-60287 analysis were performed using homogenous and isotropic material properties. Nonhomogenous effects in cable layers, like the screen and filling materials, deteriorate the analysis conditions. Therefore, small differences between the results are very obvious.

Analysis of 154 kV Three Cables in Flat Installation Using FEM

In the analysis of three cables in flat installation, the distances between the cables are considered to be twice the cable's diameter, as given in IEC-60287. Thermal and geo-

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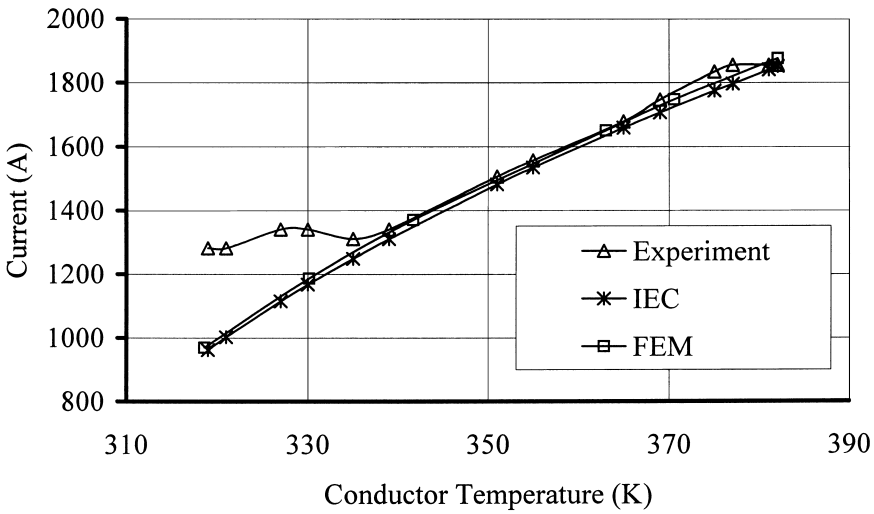

Figure 8. Conductor temperature while current increases.

metrical properties of the cable are as given in single cable layout analysis with 22 mm XLPE insulation. The cable installation is shown in Figure 9. The earth is assumed to be homogenous and isotropic for the FEM analysis.

The heat generation boundary conditions are applied at each conductor and increased gradually until the center conductor temperature reaches 90°C. The heat transfer is 44.4 W/m while the conductor temperature at the center is 90°C and at the sides are 85.08°C. Besides the thermal boundary conditions applied on the cable, the dielectric losses are also estimated ($wd = 3.57$ W/m) and applied as a thermal source over the XLPE.

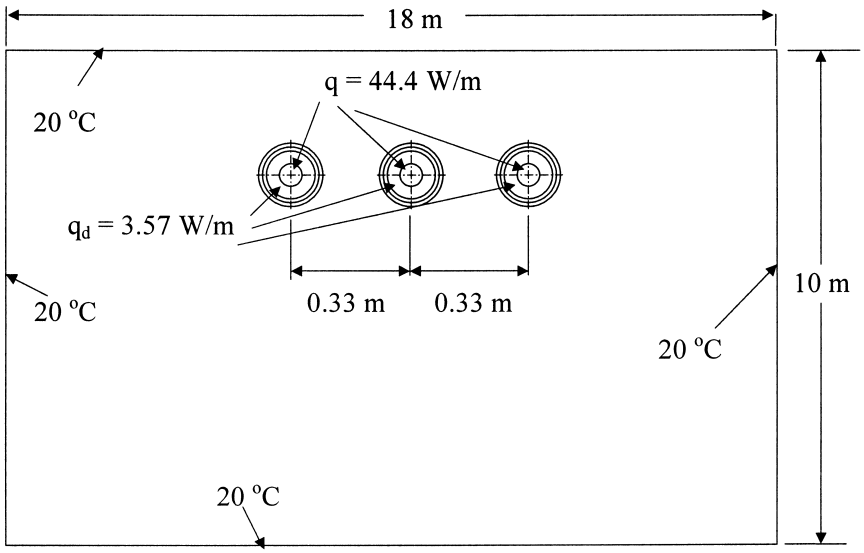

Figure 9. 154 kV cable installed in flat configuration.

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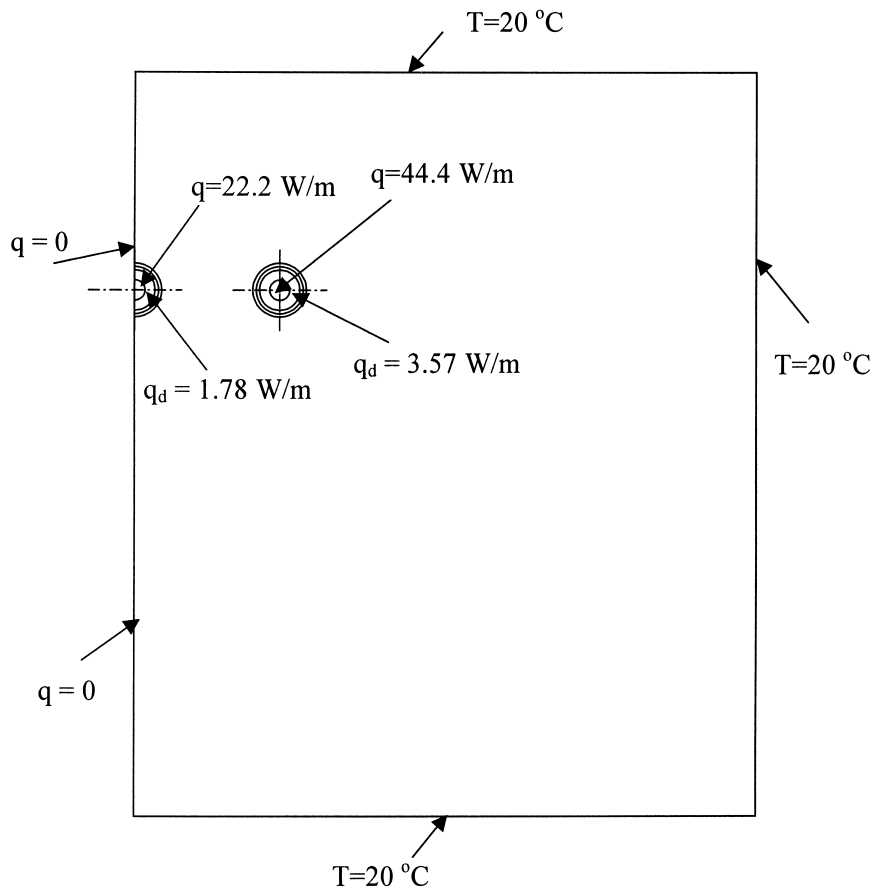

Figure 10. FEM model and boundary conditions.

Cable layout in Figure 9 shows that the domain to be considered is symmetrical in y-axis. Utilization of symmetrical boundary conditions as shown in Figure 10 on the analysis saves modeling and computation time significantly, especially in large problems. Symmetric boundary condition can be used by assuming the heat transfer on the symmetry lines are adiabatic. The mesh around the three cable installations are shown in Figures 11 and 12.

Thermal contours of the region nearby the cables are given in Figures 13 and 14. These figures illustrate that the heat transfer from the cable at the center is lower than the others. Thus, the ampacity analysis should be made for the center cable, as it is the most critical one. IEC-60287 also considers the center cable as the significant cable on its calculations. In the IEC calculations, the ampacity is 2.4% less than the numerical analysis.

Heat transfer from the cables and the ampacity values estimated using two methods are given below.

Heat transfer (IEC-60287)	$Q_{AN} = 42.35$ W/m
Heat transfer (finite element method)	$Q_{FEM} = 44.4$ W/m
Ampacity (analytical)	$I_{AN} = 1296$ A
Ampacity (finite element method)	$I_{FEM} = 1327$ A

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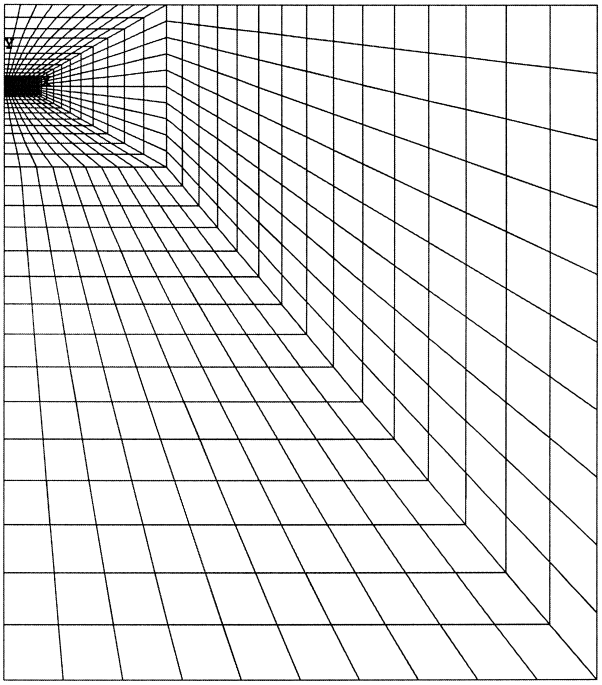

Figure 11. FEM model with 948 elements and 3001 nodes.

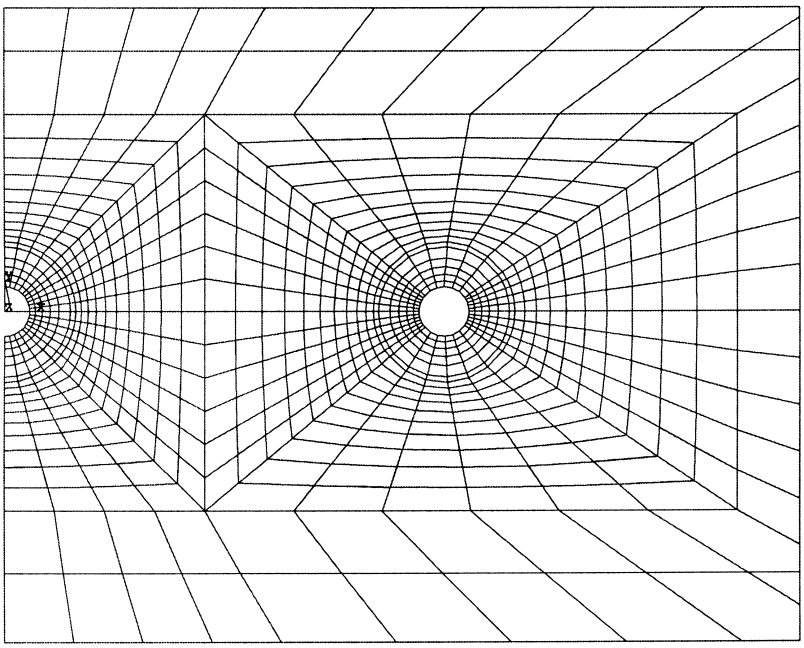

Figure 12. Mapped mesh around the cables.

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Figure 13. Temperature distribution between the cable and surrounding.

Analysis of 154 kV Cable by a Heat Source

In many cases, cable routes cross other cables or steam pipes. Due to the thermal interference in the crossing region, the cable ampacities can be decreased noticeably [13]. Analytical methods give poor results because of the geometrical complexity. However, FEM can easily analyze this kind of problem. In this numerical analysis, a heat pipe with insulation buried 600 mm below the surface is considered as shown in Figure 15. The steam temperature is normally higher than 100°C, and under steady-state conditions, the temperature on the surface of insulation is assumed to be a 60°C isotherm. Cable and pipe crossing at an angle involves a three-dimensional analysis. However, the worst case occurs when they are run in parallel in a trench. That's why the cable properties and installation dimensions are as given in the three cables installation case previously. The heat generation on the conductor is again estimated by increasing the heat generation

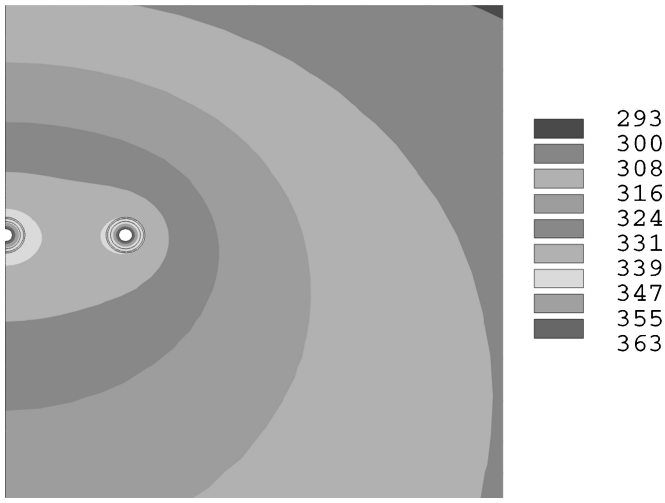

Figure 14. Temperature distribution around the cable.

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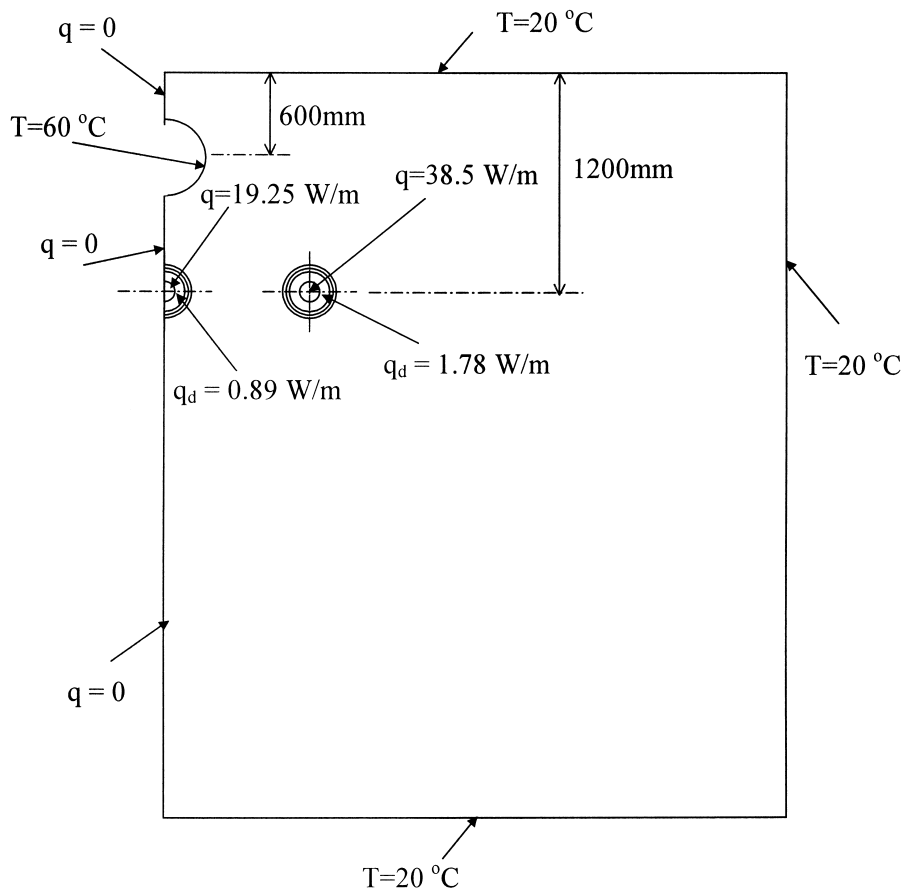

Figure 15. Cables parallel to steam pipe with boundary conditions.

gradually, until 90°C conductor temperature is reached. The mesh for FEM analysis is given in Figure 16.

When three cables are installed parallel to a heat pipe at 60 cm distance, the ampacity of the cable becomes 1236 A. Three cable installations without steam pipes have an ampacity of 1327 A, and a derating factor is necessary for application of crossing heat sources. For this case, the derating factor is 0.92. When the same boundary conditions given in the Finite Element Method section is applied to the case with steam pipe, a temperature rise of 9.4°C on the conductor is observed. Figure 17 shows the temperature distribution after the FEM analysis.

Ampacity of 154 kV cable with 22 mm XLPE insulation under different conditions are calculated by FEM as:

	FEM (A)	Analytical (A)	% difference
Single cable	1657	1635	1.3
Three cables	1327	1296	2.4
Three cables and heat source	1236	1130	9.4

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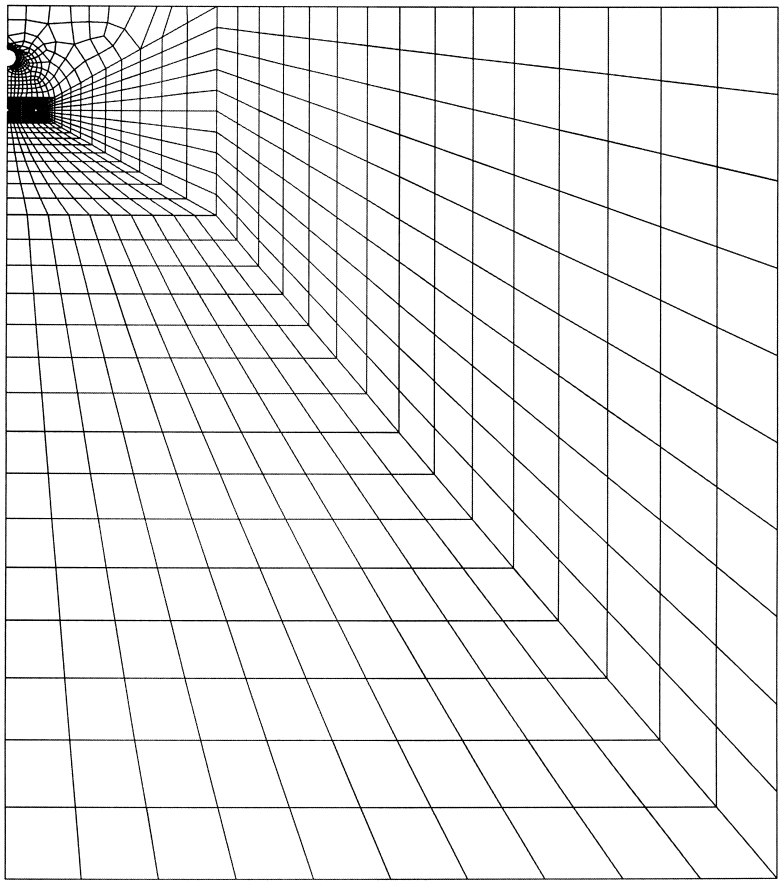

Figure 16. FEM mesh with 1215 elements.

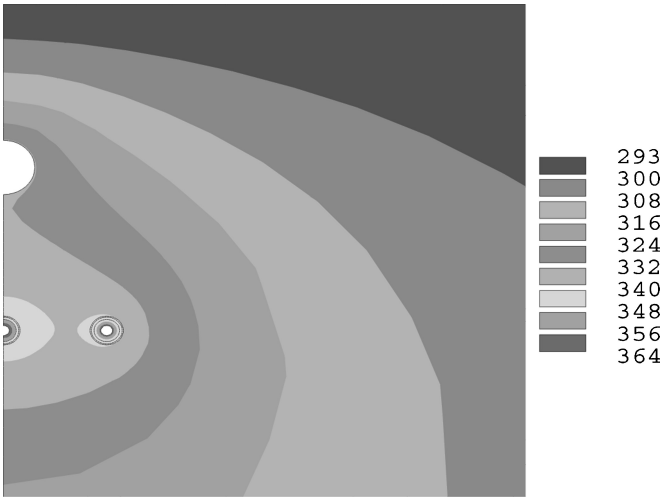

Figure 17. Temperature distribution around the cables and steam pipe.

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Conclusions

The analytical modeling of heat transfer mechanism by IEC-60287 works well in simple cable installations. However, simplifying assumptions and empirical correlation to obtain the analytical method can be significant in complex installations like crossing cable ducts, cables on trays, cables near buildings, cable splices, etc., thus making the solution become impossible. Today's computer technology gives finite element method a capability to solve any of these cases with very complex geometrical configuration. It can solve complex installations in any environment and subjected to any type of load condition together with transient analysis efficiently. When the cable surrounding is composed of various materials with different thermal resistivities, IEC-60287 formulation fails to get an acceptable result.

The numerical solutions for the underground cables showed that the finite element method gives a solution in any domain free from any geometrical complexities. Consequently, an improved calculation of ampacity saves the investment and operation expenses considerably. Currently, used power cables can be operated with higher capacity. In three cable installation analysis using FEM, the temperature distribution inside and around the cable is also determined, which is not possible using the analytical approximations. This is useful when the temperature field and specific isotherms around the cable are to be found out. The difference in the ampacity analysis using FEM and analytic solution of single cable increases in the three cable layout case.

Cables installed near the other heat sources have to be treated carefully. FEM results showed that the ampacity should be decreased by 8%. Derating factors of 0.92 found empirically could be used in crossing cable cases. However, FEM gives almost exact solutions to these kinds of problems. The mutual heating of the cables are more accurately handled in FEM.

In this study, FEM and IEC methods are assessed for some common problems. The boundary conditions and control volume determined are similar for a wide range of problems. Models for the problems with more geometric complexity can be developed using the similar approach given in this publication. Additionally the various soil thermal properties depending on the moisture content can be easily analyzed by FEM.

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